BLOOD PRESSURE & BLOOD PRESSURE DISORDERS

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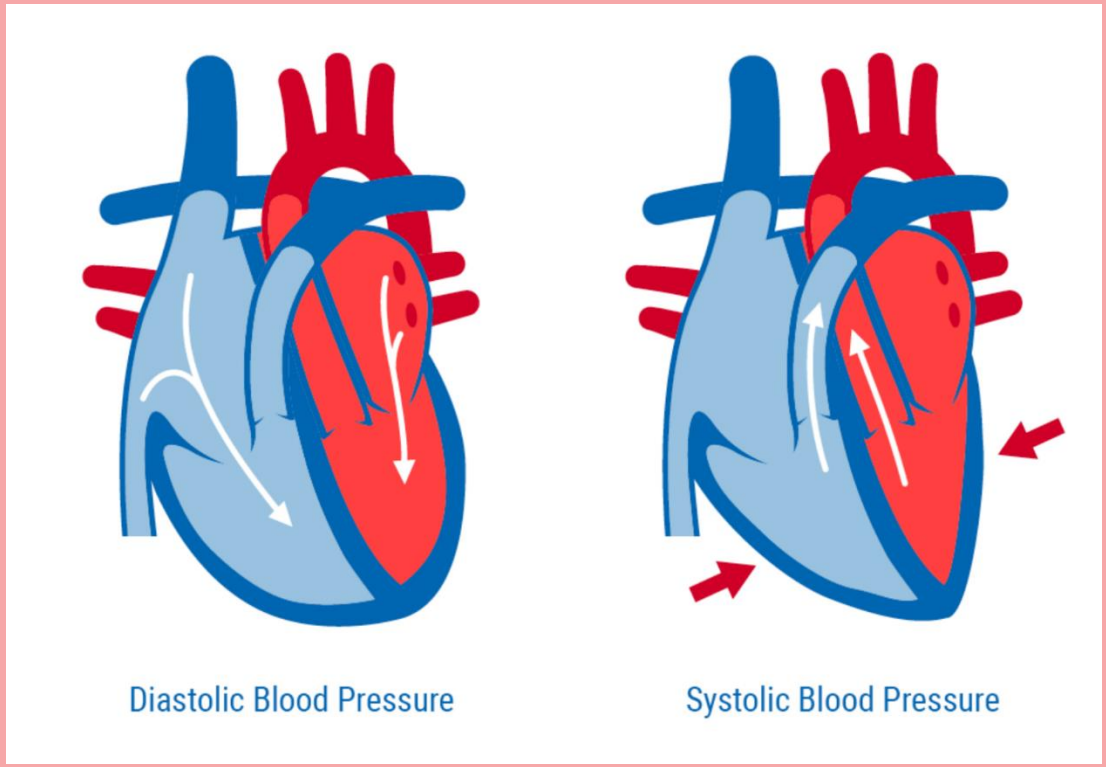


VASCULAR SMOOTH MUSCLE

- Vascular smooth muscle is located in the walls of blood vessels and regulates blood pressure and blood flow distribution throughout the body via…
  - Vasoconstriction: contraction of smooth muscle narrows blood vessel walls, increasing blood pressure.
  - Vasodilation: relaxation of smooth muscle widens blood vessel walls, decreasing blood pressure.
- Maintenance of Vascular Tone
  - VSM adjusts the diameter of blood vessels through vasoconstriction and vasodilation to regulate blood flow based on tissue needs.

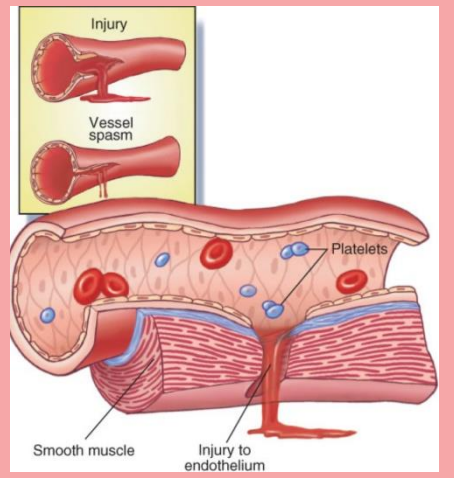
BLOOD PRESSURE TERMS

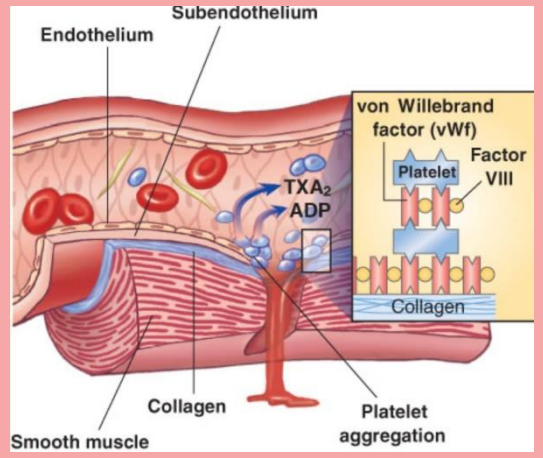
- Systolic Blood Pressure: the pressure in the arteries when the heart contracts.
- Diastolic Blood Pressure: the pressure in the arteries when the heart relaxes.
- Pulse Pressure: the difference between the systolic and diastolic blood pressure.
- Mean Arteriole Pressure: the average pressure in the arteries over a cardiac cycle. Often calculated by adding 1/3 of the pulse



VASCULAR SMOOTH MUSCLE TISSUE REPAIR

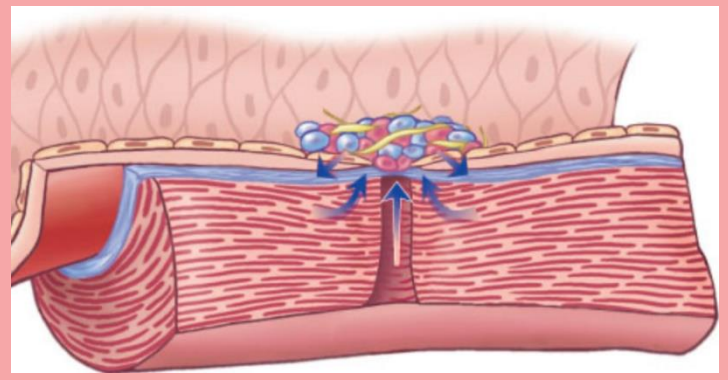
1. Vasoconstriction
   - Blood vessels constrict to reduce blood flow to the injured area.
   - Mediators released by damaged cells and platelets, such as thromboxane A2, contribute to vasoconstriction.
2. Platelet Plug Formation
   - Platelets adhere to the damaged vessel wall and become activated.
   - They release ADP and thromboxane A2 which further activates platelets and promotes aggregation.
   - Platelets form a temple plug to seal the wound.

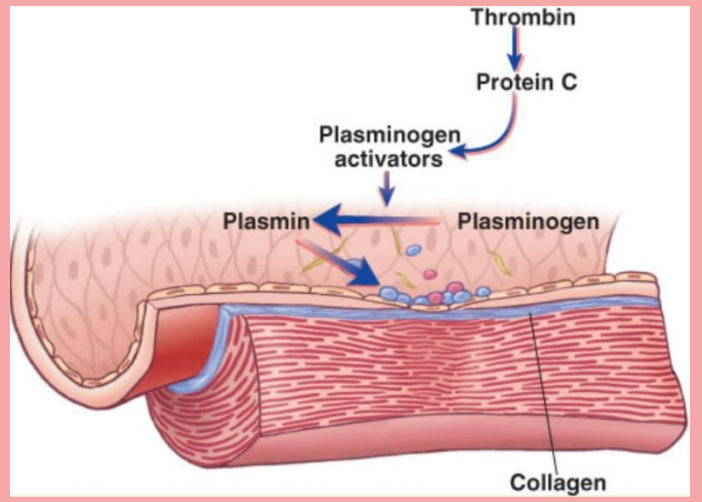




VASCULAR SMOOTH MUSCLE TISSUE REPAIR CONT…

3. Coagulation Cascade
   - Series of enzymatic reactions that convert fibrinogen into fibrin.
   - This process is initiated by the release of clotting factors from damaged cells and platelets.
   - Fibrin forms a stable cloth that reinforces the platelet plug.
4. Clot Retraction and Repair
   - The clot retracts to tighten the wound and reduce bleeding.
   - This is accomplished by the action of fibrinolysis.
   - New tissue cells eventually migrate into the clot and repair the damaged vessel.





MECHANISMS INVOLVED IN THE DEVELOPMENT OF ATHEROSCLEROSIS (BUILDUP OF PLAQUE):

- **Endothelial Injury**:
  - Damage to the endothelial cells lining the arteries is the initial step. This can be caused by hypertension, smoking, hyperlipidemia, diabetes, and inflammatory mediators.
- **Lipoprotein Deposition**:
  - Low-density lipoproteins (LDL), particularly oxidized LDL, accumulate in the intimal layer of arteries.
- **Inflammatory Response**:
  - Macrophages engulf oxidized LDL, becoming foam cells, which are characteristic of early atherosclerotic lesions (fatty streaks).
- **Smooth Muscle Cell Proliferation**:
  - Smooth muscle cells from the media migrate into the intima, proliferate, and secrete extracellular matrix (collagen and elastin), contributing to plaque formation.
- **Plaque Progression and Complications**:
  - As plaques enlarge, they narrow the arterial lumen, leading to reduced blood flow.
  - Plaques can become unstable, rupture, and trigger thrombus formation, leading to acute cardiovascular events like myocardial infarction or stroke.

WHAT IS VASCULITIS?

- Vasculitis refers to inflammation of blood vessels driven by WBC, leading to narrowing of the vessel wall
- Vasculitides are classified based on the size of the affected vessels; Large, medium, and small vessels.



IN ALL VASCULITIDES, THE AFFECTED VESSELS DO NARROW DUE TO INFLAMMATION, BUT THE EXACT MECHANISM LEADING TO THE NARROWING DIFFERS:

- **How Vessel Narrowing Occurs in Different Vasculitides:**

Large Vessel Vasculitis (e.g., Takayasu Arteritis):

- **What happens:** Long-term inflammation in the largest arteries (like the aorta) thickens and scars their inner lining, which narrows the vessel.
- **Effect:** Less blood reaches the arms or legs, causing pain when moving (limb claudication) and weak or absent pulses. If the kidney arteries are affected, it can cause high blood pressure.

Medium Vessel Vasculitis (e.g., Kawasaki Disease):

- **What happens:** Inflammation in medium-sized arteries, especially those supplying the heart (coronary arteries), makes their walls swell and thicken. Over time, scarring or blood clots from the damage can further narrow the arteries.
- **Effect:** Reduced blood flow to the heart, which can lead to chest pain (ischemia), heart attacks (myocardial infarctions), or bulges in the artery (aneurysms). After healing, scarring can cause lasting narrowing.

Small Vessel Vasculitis (e.g., Henoch-Schönlein Purpura - HSP):

- **What happens:** The immune system deposits harmful proteins (IgA) in tiny blood vessels, triggering inflammation. This causes swelling and tiny clots that block blood flow.
- **Effect:** Reduced blood supply causes:
    - **Purple skin spots (palpable purpura)** from damaged skin vessels
    - **Belly pain** from low blood flow to the intestines
    - **Kidney problems (nephropathy)** from damaged kidney filters

AORTIC ANEURYSM

Definition:

<u>True aneurysm- B</u>ounded by complete vessel wall.

- Blood remains within vascular compartment.
- Dilation of all three vessel layers.

False aneurysm- Localized dissection or tear in artery inner wall.

Formation extravascular hematoma.

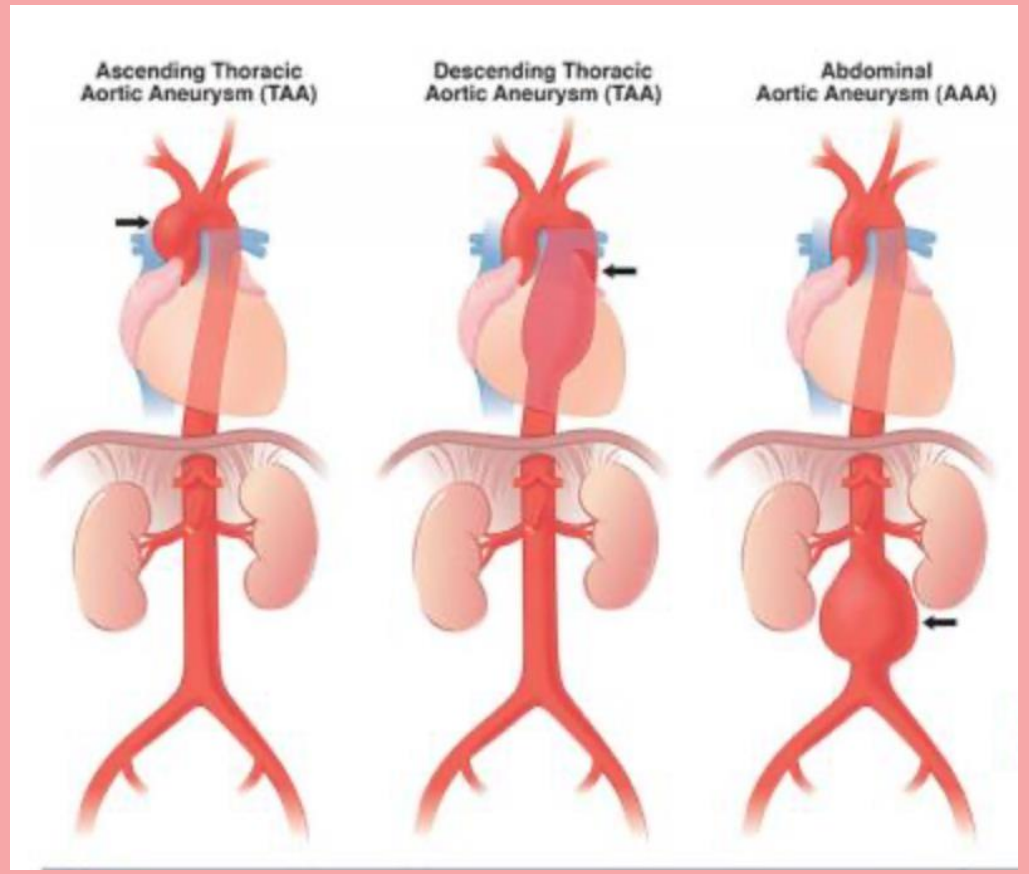
Outer layer of the vessel wall is only supporting layer.

Pathogenesis:

- Aortic aneurysm is a bulge located in the wall of the aorta. This bulge can occur anywhere within the length in the aorta. If the bulge grows too large, the arterial wall will burst leading to internal bleeding.

- S/sx:
- -Can present asymptomatic
- -Pulsating sensation in abdomen
- -Deep, constant pain, in abdomen or back

- Risk factors include:
- 65 years>
- Smoking
- HTN
- Fam HX
- Atherosclerosis
- Trauma

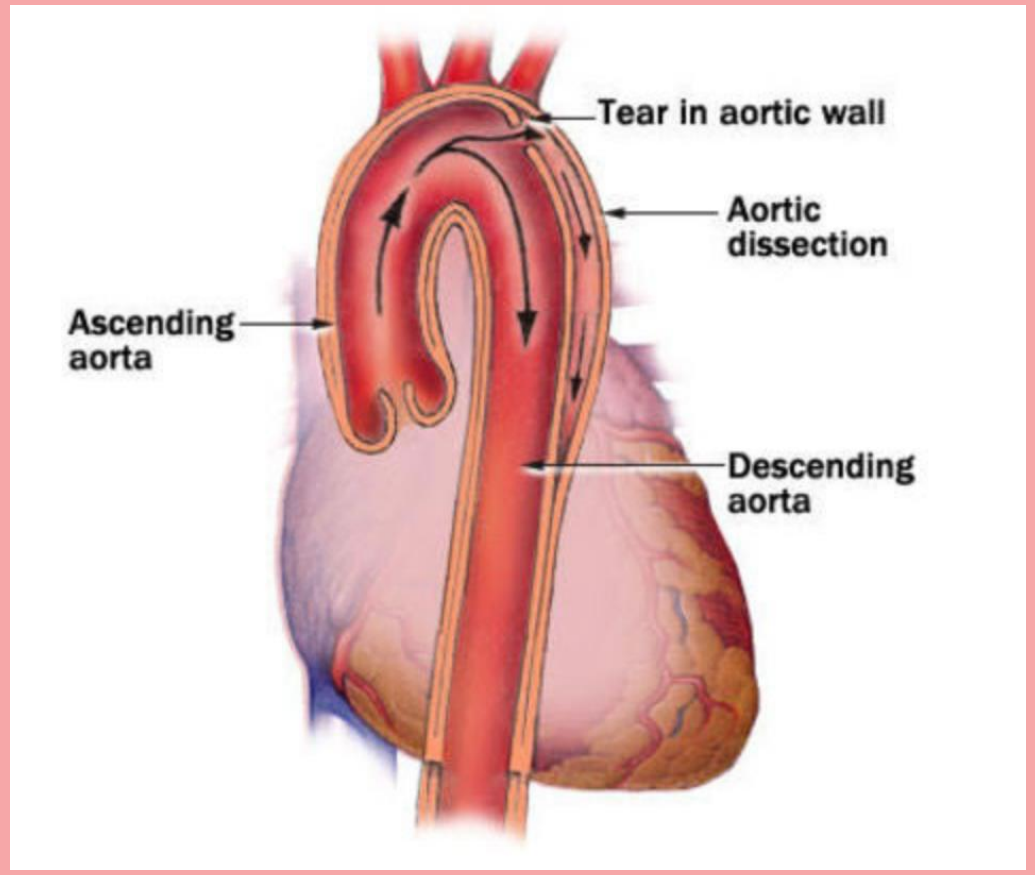


AORTIC DISSECTION

Pathogenesis:
- Tear within the inner layer of aorta.
- Tear creates false lumen, resulting in blood flow. between layers of the aortic wall.
- Acute life- threatening condition.

Signs and Symptoms:
- Sudden, severe chest pain/back pain.
- Tearing or ripping sensation.
- SOB.
- Loss of consciousness.

Risk factors:
- HTN.
- Most common in 40-60 year old men.
- Connective tissue diseases (Marfan Syndrome).
- Surgery- aorta incised or cross clamped.
- Congentital aortic valve abnormalities, coarctation.



DIAGNOSIS

Physical exam:

- Pulsatile abdominal mass.

Imaging:

CT Angiography: The gold standard.

- MRI- Useful, especially in stable pts.
- Transesophageal Echocardiography (TEE): Can be performed if other imaging is not available.
- Ultrasound.

Treatment:

- Medical management:
- Control blood pressure (beta blockers) reduce stress on aorta.
- Control cholesterol through medication (statins) and life-style changes (diet).
- Pain management.

Surgical Intervention:

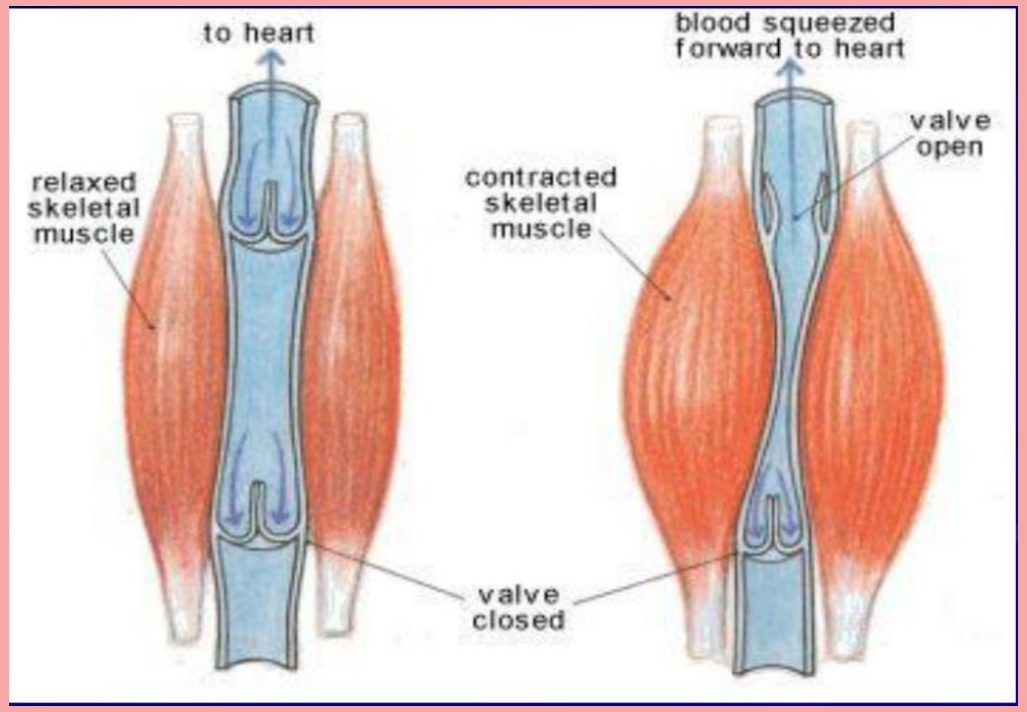
- Endovascular stenting, open surgery

VENOUS RETURN FROM LOWER EXTREMITIES

- Blood is transported back to the heart from the lower extremities.

- **Mechanisms:**
- Muscle Pumps- Skeletal muscle contractions compress veins, propelling blood upward.
- One-way valves prevents backflow during muscle relaxation.

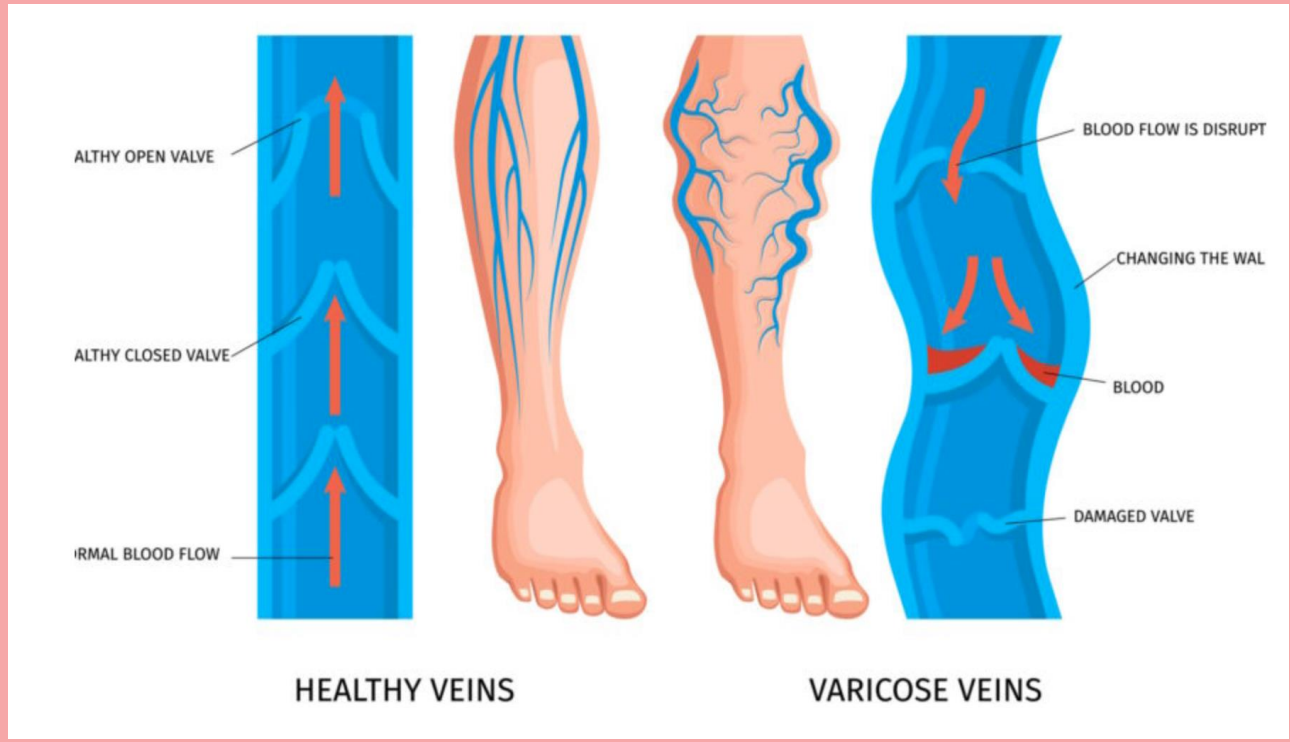
- **Effects of Gravity:**
- Gravity challenges venous return when standing or sitting.
- Muscle pumps and valves counteract gravitational effects.



VARICOSE VEINS

- **Pathogenesis:**
- Enlarged, twisted veins that are often dark blue/purple.
- Typically located in legs and feet
- Result from weakened or damaged valves within veins, leading to poor blood flow and pooling of blood.

- **Causes:**
- Valve dysfunction leads to backflow and increased vein pressure.
- Weak vein walls due to aging, pregnancy, obesity, or prolonged stationary standing.



DIAGNOSIS AND TREATMENT

- **Diagnosis:**
- Med hx: Review symptoms, family history, and risk factors.
- Physical Examination: Inspections of legs for swollen viens and skin changes.
- Ultrasound: Non-invasive imaging to assess blood flow and check for valve function.
- **Treatment:**
- Life-style changes.
- Compression stockings.
- Sclerotherapy-Injection of solution to close off affected veins.
- Endovenous Laser Treatment- Lasey energy used to seal off varicose veins.
- Vein stripping- Surgical removal of larger varicose veins.
- Ambulatory Phlebectomy- Removal of smaller varicose veins through tiny incisions.

WHAT IS HYPERTENSION?

Hypertension- sustained elevation of blood pressure within the arterial circuit

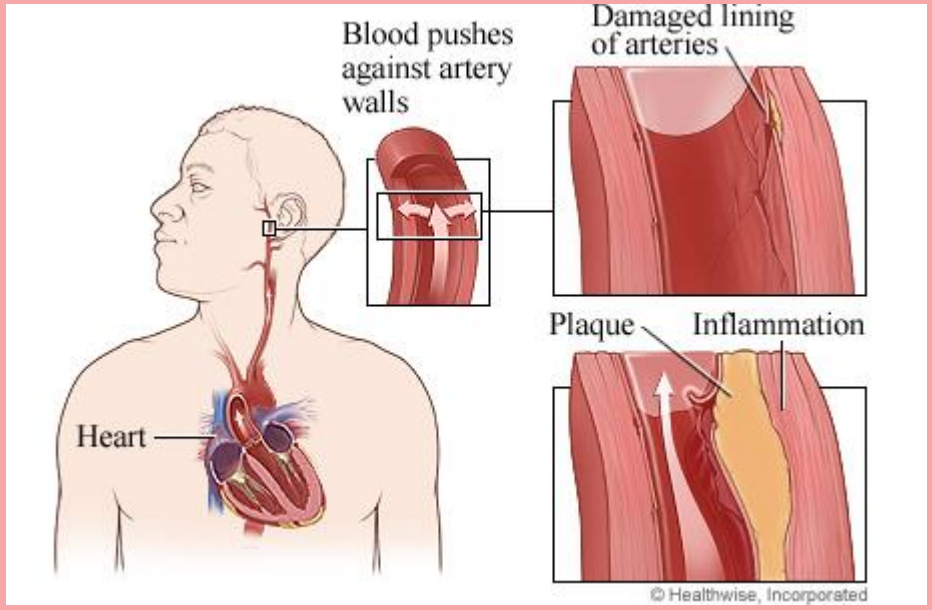
- Normal Blood Pressure:
  - Systolic: <120
  - Diastolic: <80
- Elevated Blood Pressure:
  - Systolic: 120-129
  - Diastolic: <80
- Stage 1 Hypertension
  - Systolic: 130-139
  - Diastolic: 80-89
- Stage 2 Hypertension:
  - Systolic: >140
  - Diastolic: >90
- Hypertensive Emergency:
  - Systolic: >180
  - Diastolic: >120

TYPES OF HYPERTENSION

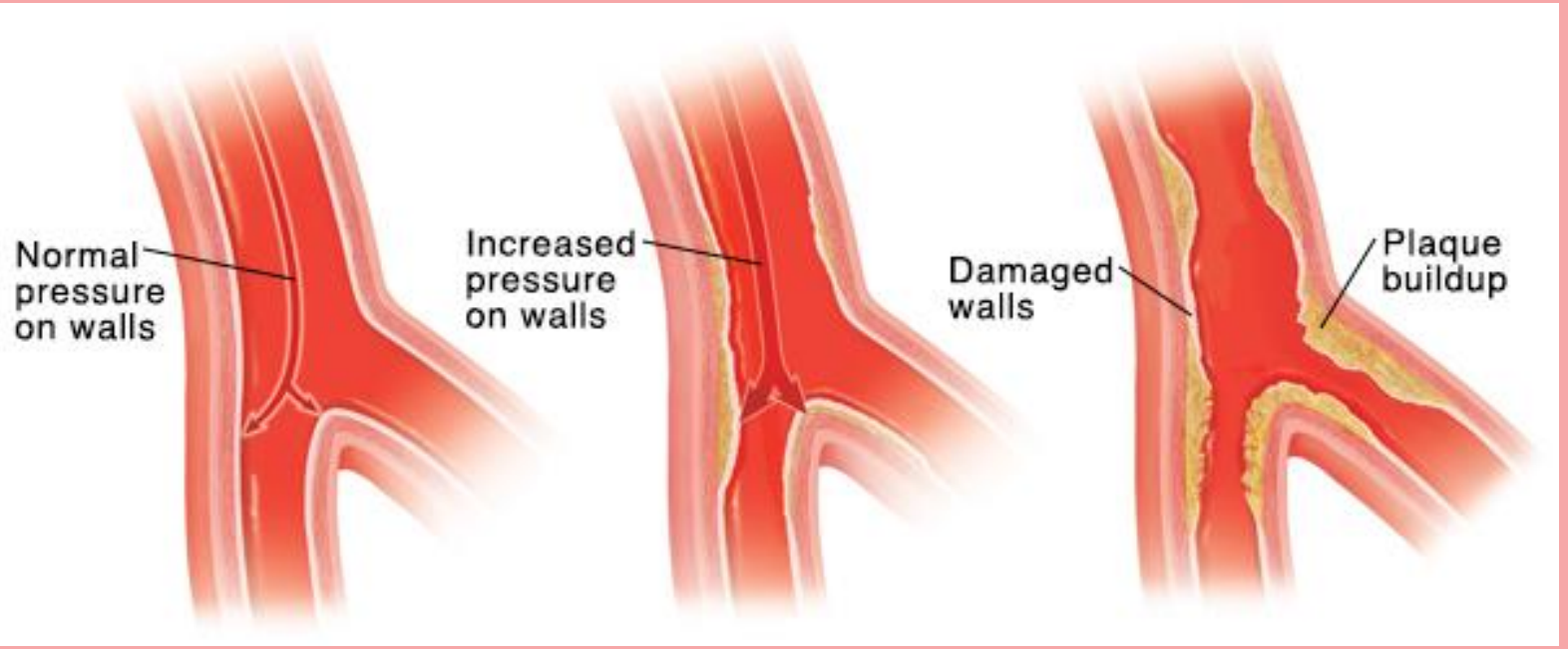
- Primary Hypertension: clinical presence of hypertension without evidence of causative clinical condition
  - Non-modifiable risk factors: age, gender, race, family history, and genetics
  - Modifiable risk factors: diet, levels of blood lipids, tobacco/alcohol consumption, activity level, obesity, and blood glucose control
- Secondary Hypertension: elevation of blood pressure due to another disease condition
  - Kidney disease
  - Disorders of adrenal cortical hormones
  - Pheochromocytoma
  - Coarctation of Aorta
  - Oral contraceptive use

TARGET-ORGAN DAMAGE

- Increased perfusion pressure can damage organs
- Increased intravascular pressure can damage vascular endothelial cells which can increase the risk for the plaque build-up in the arteries (atherosclerosis) or vascular rupture
- Increased blood pressure consistently puts excessive pressure (or force) on the walls of blood vessels causing them to narrow, weaken, and become damaged
- Target organs: highly vascularized or dependent of adequate blood supply for their function



DAMAGE TO WALLS OF BLOOD VESSELS



ORGANS AFFECTED IN TARGET ORGAN DAMAGE



Heart

Coronary heart disease

Cardiac dysrhythmias

Myocardial infarction

Heart failure

Sudden death



Vascular System

Atherosclerotic Vascular Disease

Peripheral Artery Disease



Brain

Stroke or transient ischemic attack



Kidneys

Chronic kidney disease

Kidney failure



Eyes

Retinopathy

HYPERTENSION IN PREGNANCY

Preeclampsia-Eclampsia	Chronic Hypertension	Chronic Hypertension with Superimposed Preeclampsia	Gestational Hypertension
• Hypertension • Thrombocytopenia • Impaired liver and renal function • Pulmonary edema • Vision and cerebral disturbances • "Eclampsia" phase • HELLP syndrome	• Hypertension present before pregnancy or before 20 weeks	• Hypertension present before pregnancy or before 20 weeks • Preeclampsia symptoms present after 20 weeks: markedly elevated BP and protein in the urine	• Hypertension develops after 20 weeks' gestation • No evidence of preeclampsia • BP normalizes after delivery

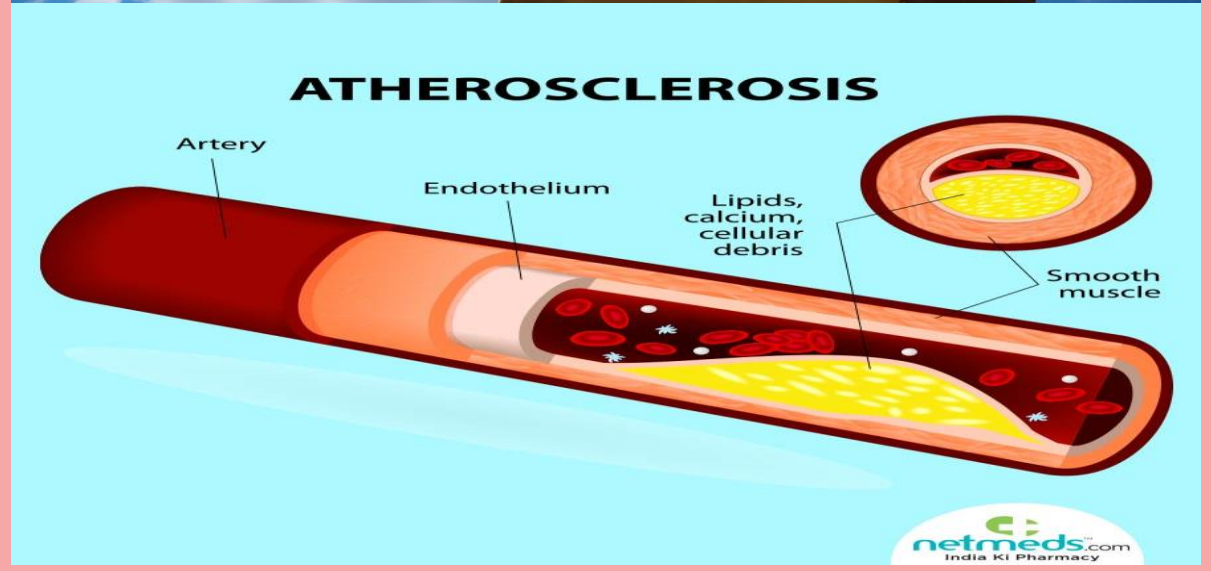
ORTHOSTATIC HYPOTENSION

- Orthostatic hypotension is defined as 20mm Hg decrease in Systolic bp and 10 mm Hg decrease diastolic within 3mins of standing from the supine position.
- Orthostatic Hypotension mostly occurs in elderly population (65 and above).
- Risk factors: Age, medications (ACE inhibitors and Ca channel blocker), Dehydration, Cardiovascular diseases.
- Persistent Orthostatic Hypotension can cause Fall, Stroke, and Heart failure

CAUSE OF ORTHOSTATIC HYPOTENSION

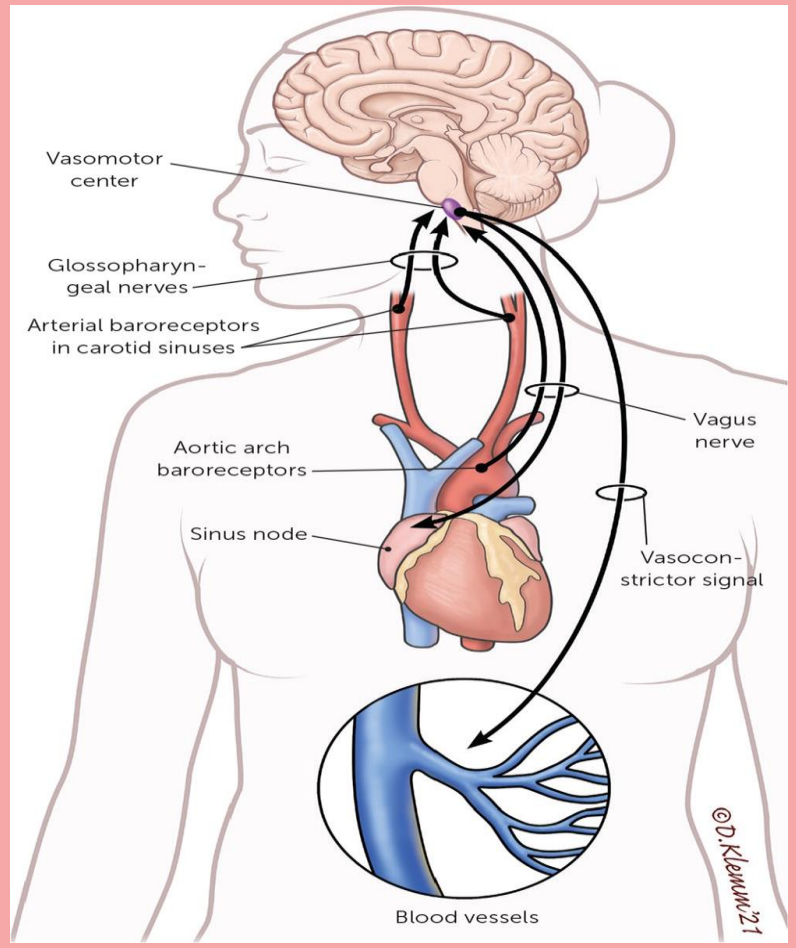
- Age- Blood vessels loses its elasticity with age.
- Dehydartion/Water deficit
- Medications such as ACE, ARB, diuretic, Ca channel blocker can cause orthostatuic Hypotension
- Cardiovascular disease and Atherosclerosis
- Autonomic Nervous system defects; Baroreceptors defects





BODY RESPONSE TO ORTHOSTATIC HYPOTENSION

- Baroreceptors in the aortic sinus and carotid arch activates the CNS.
- Sympathetic System activation increase the HR and contractility.
- Increasing CO and vasomotor tone in the capacitance vessel
- Vasopressin, ADH is released to reabsorb water from the kidney.
- Renin, Aldosterone and Angiotensin is released to retain sodium and water.

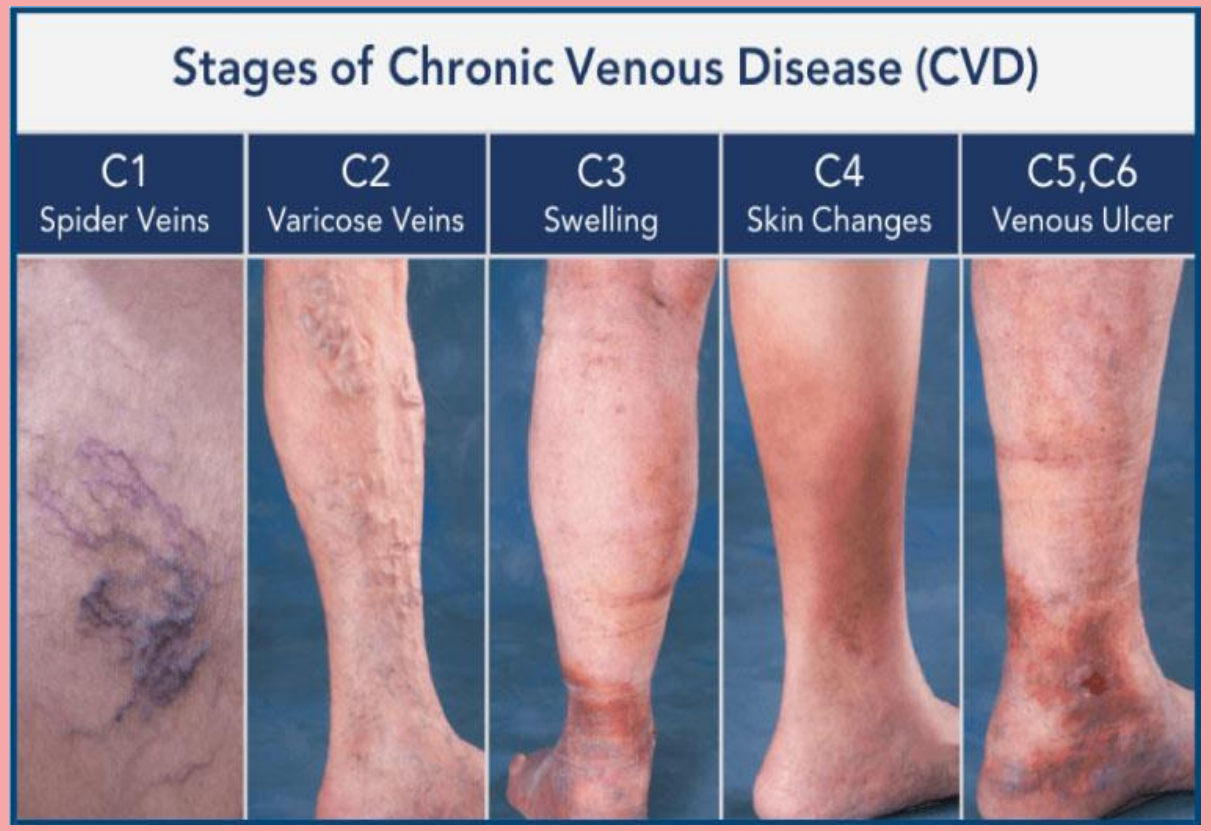


VENOUS INSUFFICIENCY

Definition: A condition in which the veins, are unable to return blood to the heart due to dysfunction of the venous valves

Pathology:

- Valve dysfunction and venous reflux
- Venous Hypertension and Vein Dilation
- Capillary Leak and Tissue Damage
- Chronic Inflammation



Actions to take:

Treatment typically focuses on improving venous circulation, reducing edema, and addressing the underlying venous insufficiency through compression therapy, lifestyle changes, or surgical interventions.

RELATION TO STASIS DERMATITIS AND VENOUS ULCERS

Stasis Dermatitis:

Common skin condition that is associated with chronic venous insufficiency. Results from hypertension, venous reflux, and edema. The constant high pressure in the veins leads to capillary leakage which blood components leak out into the surrounding tissues which cause inflammation.

Signs and Symptoms:

- Dry skin that becomes inflamed, red, itchy.
- Skin may become thicker/hard
- May cause skin ulcers

Venous Ulcers:

A severe complication of chronic venous insufficiency and they usually happen due to prolonged venous hypertension, capillary leaking and chronic inflammation. The elevated venous pressure prevents adequate blood flow depriving the tissues of oxygen and nutrients which are essential for tissue repair and normal function. The skin then becomes less vulnerable and will start to breakdown and ulcers can form.

Location and Appearance:

- Most commonly near the ankles
- Typically appear as shallow, irregularly shaped wounds with a yellow or sloughy base
- Has undermined edges and may be significant drainage from the ulcer

MOST COMMON CAUSES OF LEG ULCERS

1. Varicose veins → prolonged distension and stretching of the superficial veins due to venous insufficiency.
2. Varicosities → defects in the superficial veins or because of impaired blood flow in the deep venous channels.
3. Chronic Venous insufficiency → usually due to a type of valvular incontinence
4. Venous thrombosis → thrombus in a vein and the accompanying inflammatory response in the vessel wall.

FUNCTIONS OF ENDOTHELIAL CELLS

- Endothelial cells- specialized cell that lines the inside of blood vessels, lymphatic vessels, and the heart. This forms a single layer called the endothelium.
- Blood pressure regulation-by producing substances like nitric oxide, that cause vasodilation, they help control BP and blood flow throughout the body.
- Barrier function- form a selective barrier, managing what substances can pass through, maintaining tissue homeostasis.
- Angiogenesis- this is the development of new blood vessels.
- Immune cell trafficking- they express adhesion molecules that allow immune cells to adhere to the vessel wall and migrate into tissues when needed.
- Hemostasis- they release factors that promote platelet aggregation and fibrin formation at the sites of injury, managing blood clotting.
- Fluid filtration- they regulate the movement of fluid and soultes between the blood and surrounding tissues through their tight junctions
- Endothelial dysfunction- this is a condition where the lining of the blood vessels(endothelium) becomes impaired leading to abnormal blood vessel function.

CARDIAC OUTPUT & PERIPHERAL VASCULAR RESISTANCE

- Cardiac output- the amount of blood pumped by the heart per minute, this is the product of stroke volume and heart rate.
- Peripheral vascular resistance-narrowing of the blood vessels/the collective resistance of all vessels.
- Cardiac output and peripheral vascular resistance directly interact to determine blood pressure, where a higher cardiac output, combined with increased peripheral vascular resistance leads to elevated systolic and diastolic blood pressure, while the opposite scenerio results in lower blood pressure.
- Diastolic pressure is primarily affected by peripheral vascular resistance and systolic pressure is largely affected by cardiac output.